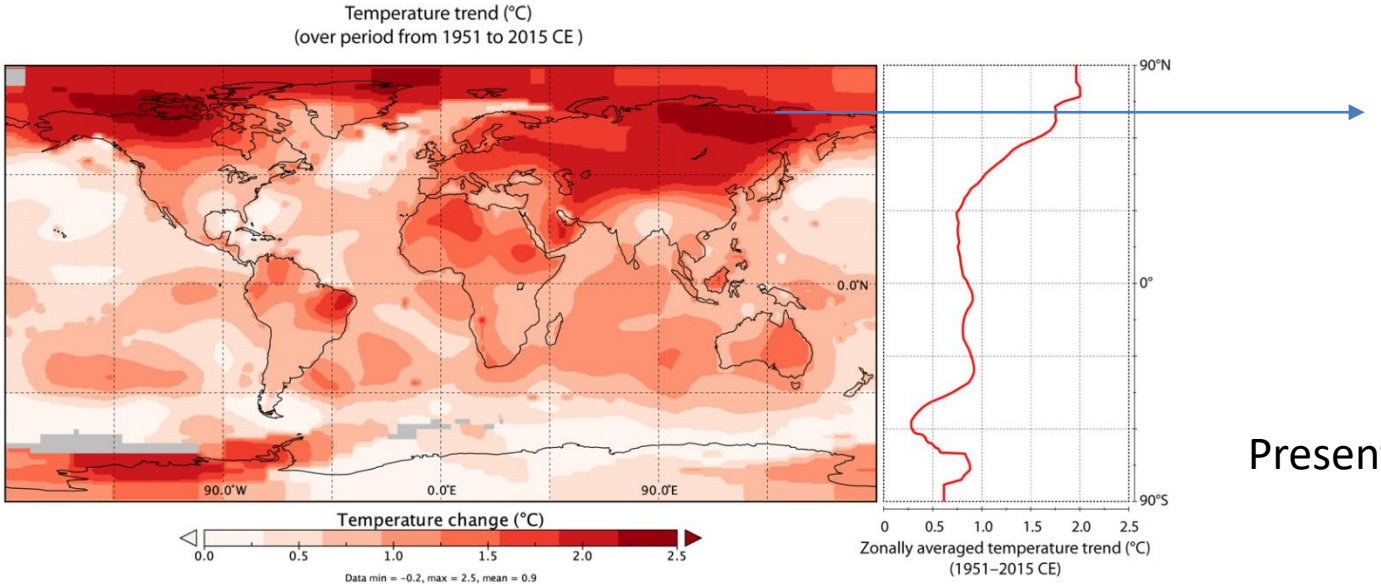


Polar Amplification

Reason?

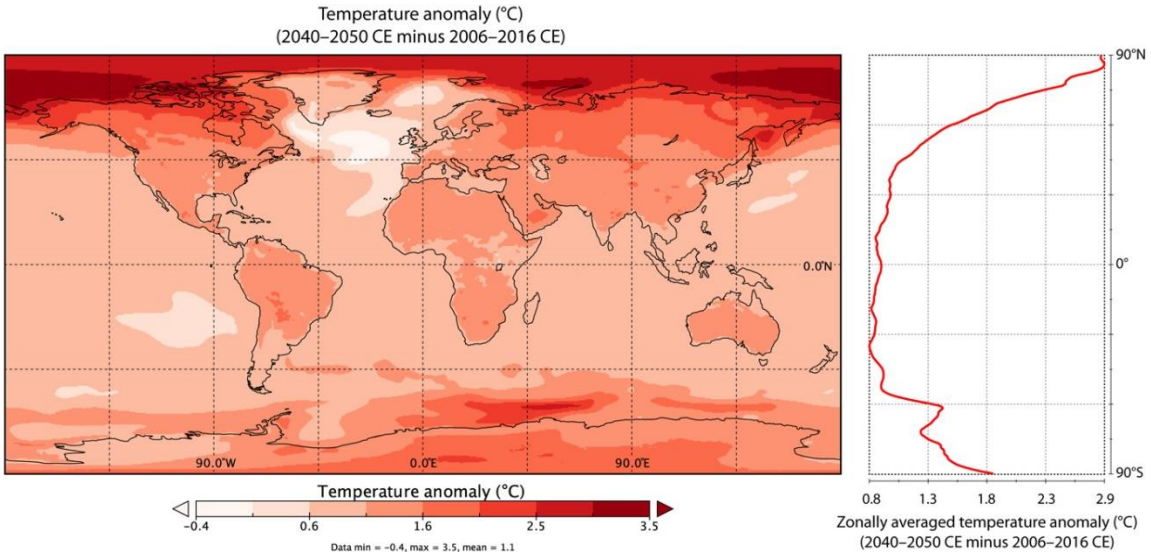
Surface Albedo Change

Present

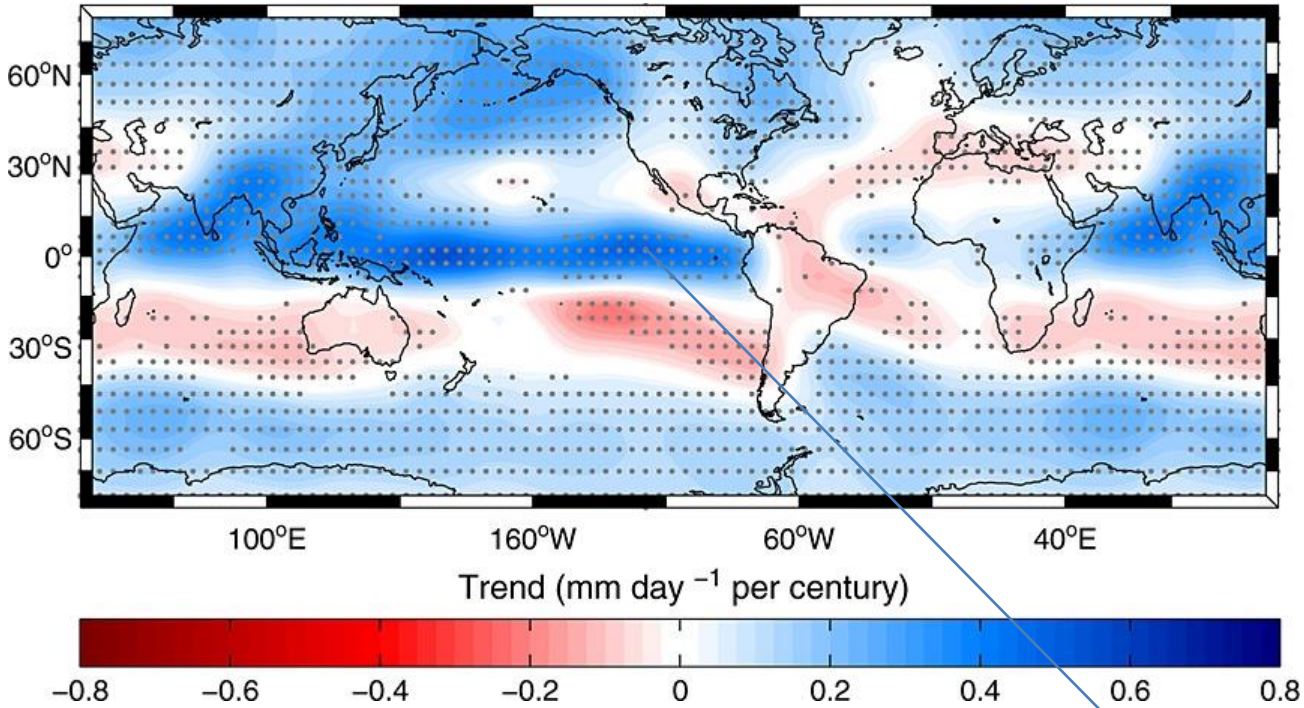
Spatial Pattern of Surface Air Temperature Changes

Putnam et al., 2017, Science Advances

Future



Ren et al., 2013, JGR



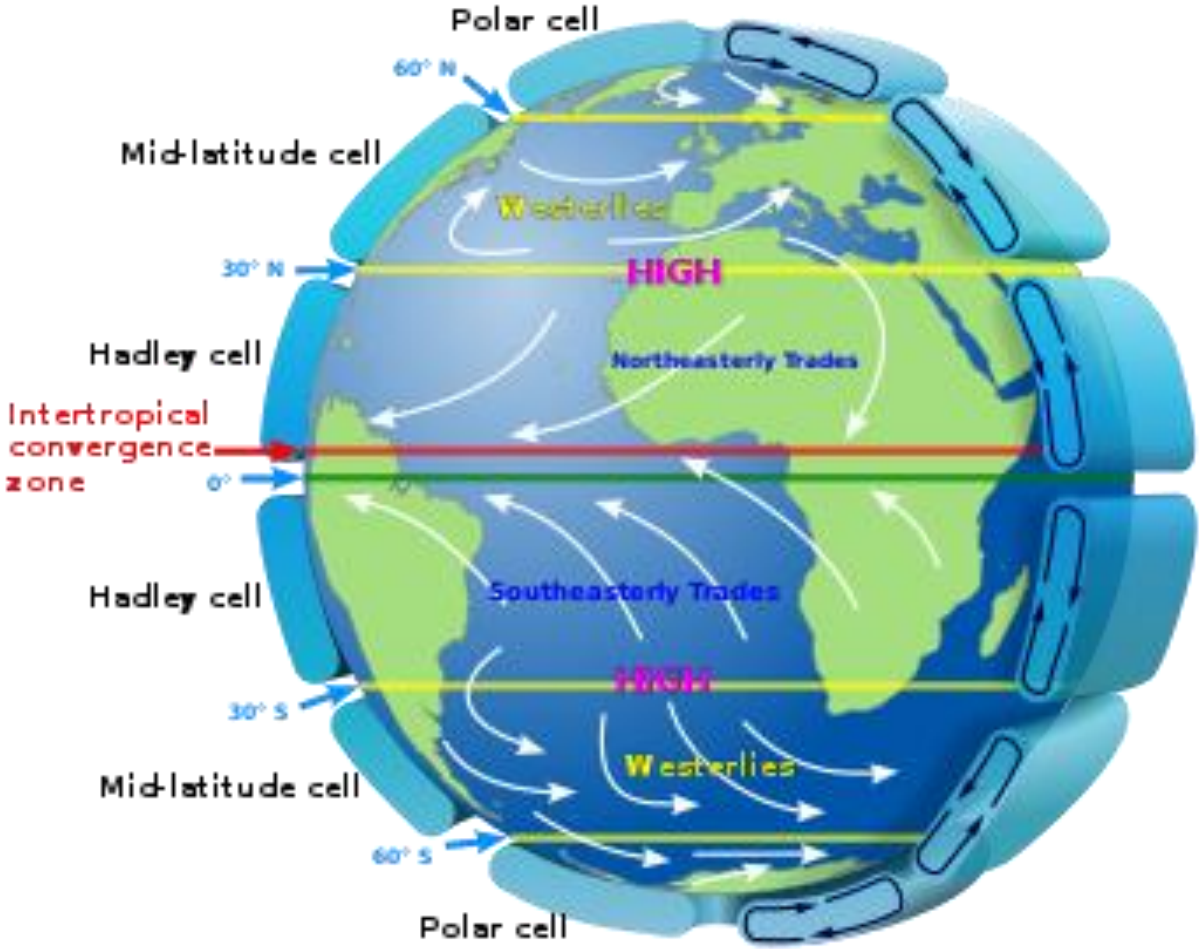
Rainfall changes due to climate change?

Increase in the equatorial region

Decline in the sub-tropics

Reason

Slowing atmospheric circulation

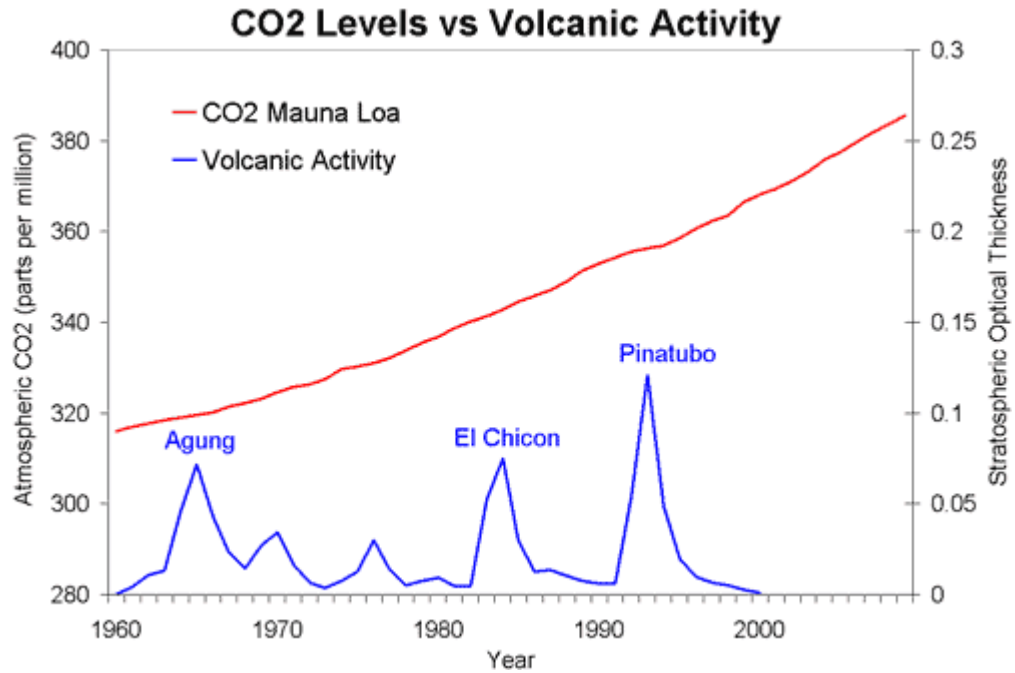


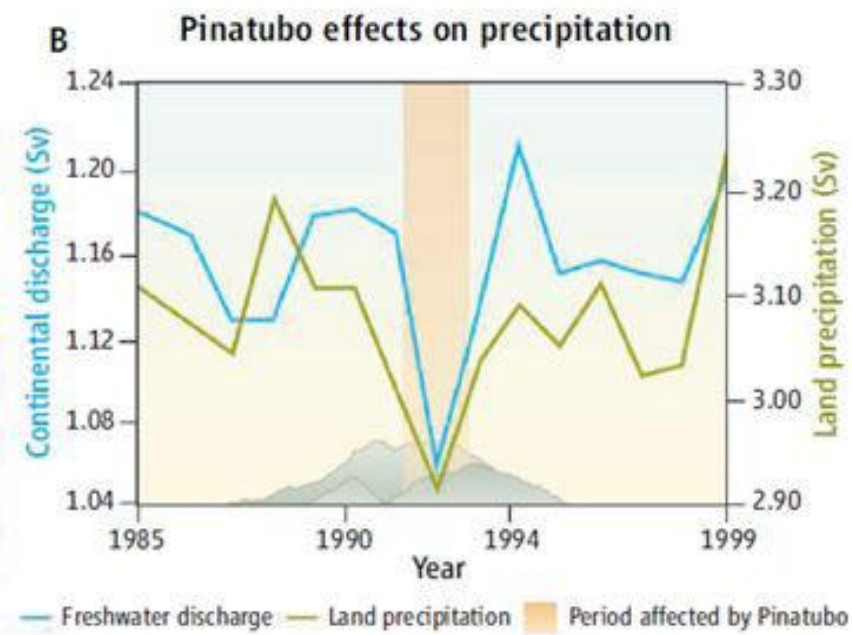
Climate change and its impact is a multi-faceted issue. We only covered global warming.

You may read the rest if interested.

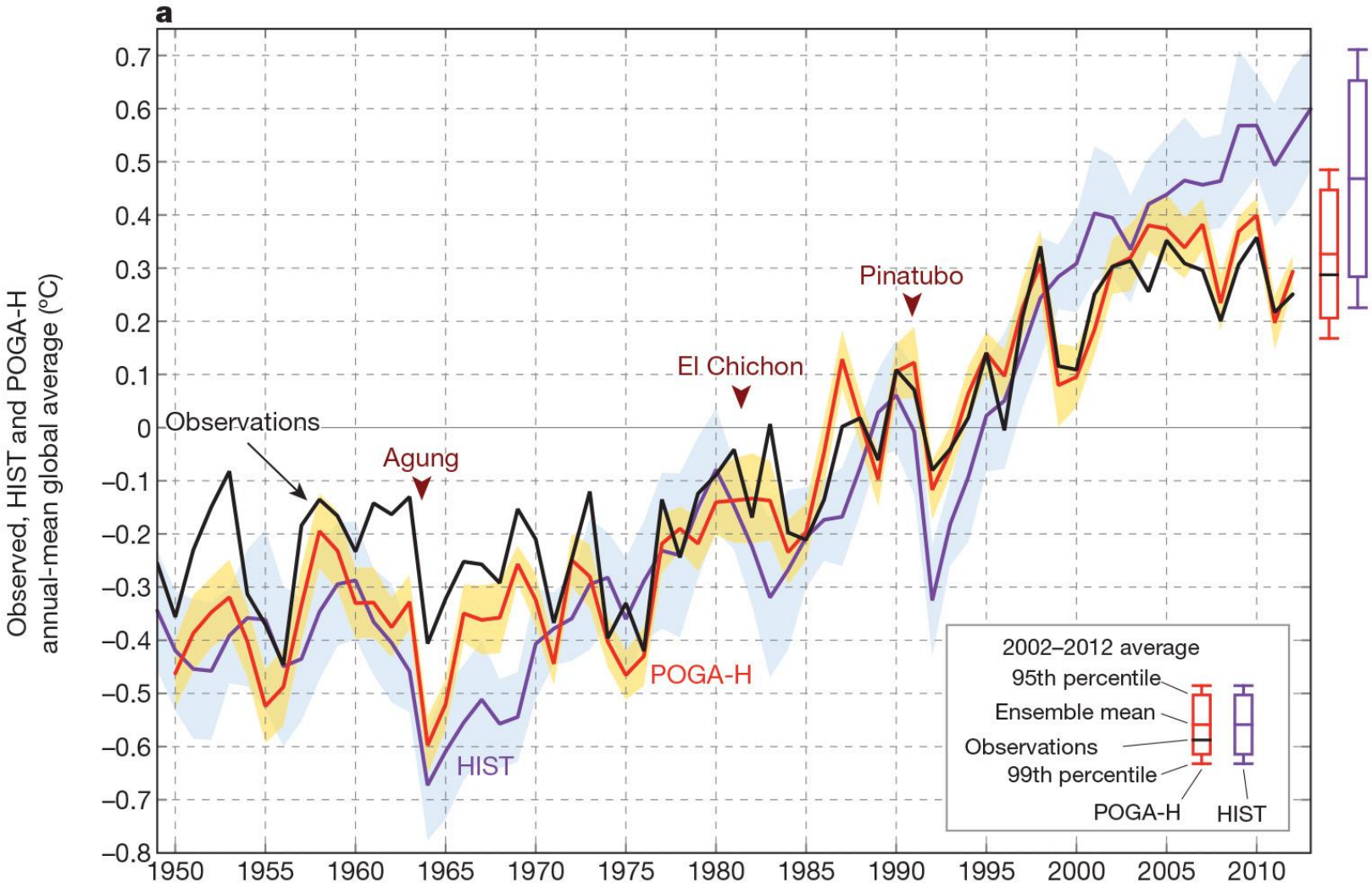
CO2 level increase and Volcanic Activity are related, but is not the sole cause for the trends

Look at the picture to the right…





Volcanic Emissions have a significant effect on rainfall. Surface temperature changes may persist for a few years.

Look at the picture to the left…

Volcanic Eruption and Global Mean Temperatures